

PROFILE

I'm a Blockchain developer specialising in **Solidity Smart Contract** development with a focus on **DeFi Protocols**, on-chain automation, and rigorous testing. Experienced in deploying and verifying production contracts on Ethereum testnets, integrating **Chainlink Oracle**, and implementing AMM mechanics. Grounded in **SC Security** principles and advancing into **Zero-Knowledge Cryptography** as a long-term specialisation.

SKILLS

SMART CONTRACTS

SolidityFoundryDeFiChainlink OracleOpenZeppelinSepolia

SECURITY & TESTING

Fuzz / Unit TestingReentrancyGuardAccess ControlSC Security Basics

TOOLING & LANGUAGES

RustLinuxGitForge / Cast / AnvilEtherscan

ZK / CRYPTOGRAPHY

NoirSNARKs / STARKsZK Rollups

PROJECTS

Decentralized Provably Fair Raffle

github.com/04arush/Raffle-Contest

Solidity • Chainlink VRF • Chainlink Automation • Foundry

On-chain raffle with cryptographically verifiable randomness via Chainlink VRF v2.5 and fully autonomous winner selection using Chainlink Automation — no admin intervention required. Features a chain-aware deployment syste (HelperConfig pattern) supporting both Anvil local mocks and Sepolia live contracts.

AMM Decentralized Exchange (DEX)

github.com/04arush/AMM-DEX

Solidity • OpenZeppelin • ERC-20 • Foundry

Constant-product AMM ($x \cdot y = k$) with two custom ERC-20 tokens, LP token minting, 0.3% swap fee, slippage protection, and reentrancy guards. Includes a web3 frontend for live interaction. 10 tests passing including fuzz tests.

Employee Payroll Manager

github.com/04arush/Employee-Payroll-Manager

Solidity • Chainlink Automation • OpenZeppelin • Foundry

Automated payroll system with on-chain employee records, configurable payment frequencies, and Chainlink Automation for trustless salary disbursement. Secured with ReentrancyGuard and Ownable access control..

EDUCATION

Indira Gandhi National Open University (IGNOU)

Aug 2023 - Ongoing

Bachelor's — Computer Application

CERTIFICATIONS

Noir Programming & ZK Circuits	Cyfrin Updraft • In Progress
Fundamentals of Zero-Knowledge Proofs	Cyfrin Updraft • Mar 2026
Advanced Foundry	Cyfrin Updraft • Mar 2026
Foundry Fundamentals	Cyfrin Updraft • Mar 2026
Solidity Smart Contract Development	Cyfrin Updraft • Jan 2026
Rust Programming Basics	Cyfrin Updraft • Feb 2026
Advanced Web3 Wallet Security	Cyfrin Updraft • Feb 2026
Blockchain Basics	Cyfrin Updraft • Dec 2025